

Relationships Between Harvest Indices and other Plant Characteristics in Soybeans¹

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ABSTRACT

Indeterminate, semideterminate, and determinate soybean [*Glycine max* (L.) Merr.] strains in maturity groups I, II, III, and IV were grown to examine relationships among actual harvest index (seed yield/above-ground biomass), apparent harvest index (seed yield/above-ground mature plant weight), seed yield, plant height, maturity, and other plant characteristics.

Highly significant correlations between the actual and apparent harvest index means indicated that the apparent harvest index adequately reflected the actual harvest index of a genotype. Therefore, measurement of the apparent harvest index should permit a valid comparison of the relative efficiency of a set of genotypes.

The actual and apparent harvest indices were inversely correlated with biological yield (above-ground biomass), vegetative yield (above-ground non-grain biomass), the components of vegetative yield (straw weight and leaf weight), maturity, and plant height. The latter six characteristics were positively correlated. Harvest index and seed yield were not consistently related from one season to the next. No relationship was detected between the growth habit and the harvest index of a genotype.

Correlations between the 1975 and 1976 means revealed that the relative performance of the entries based on harvest index means was less erratic than the relative performance based on seed yield.

Additional index words: *Glycine max* (L.) Merrill, Biological yield, Phenotypic stability, Genetic variability.

HARVEST index has been defined in a number of ways depending upon the particular experiment and plant species studied. Donald (6) originally defined harvest index as the ratio of seed yield to biological yield, with biological yield denoting the total above-ground dry matter produced by the plant. Experiments examining harvest index in soybeans [*Glycine max* (L.) Merr.] have defined this index as the ratio of seed yield to mature plant weight (3, 14). For some plant species, these two ratios would be quite similar or perhaps equivalent, because the difference between biological yield and mature plant weight would be insignificant. For soybean genotypes, however, these ratios are distinctly different. To differentiate between these two ratios the efficiency index, based on seed yield and biological yield, will be referred to as the actual harvest index, and the index, based on seed yield and mature plant weight, will be referred to as the apparent harvest index. When the distinction is unnecessary they will be collectively referred to as harvest index.

Donald (6) emphasized the possible importance of harvest index and biological yield and suggested utiliz-

ing these two characteristics to improve the efficiency of breeding for seed yield. By identifying the most efficient genotypes, and those with superior biological and seed yielding abilities, crosses could be made to attempt to combine these characteristics into one genotype. Such an approach may prove beneficial, because one or both of these components must be modified to increase seed yield (10, 12, 13).

However, implementing this type of approach is difficult in a soybean breeding program. Estimates of biological yield and actual harvest index for a soybean genotype cannot be calculated unless leaf blades and petioles are collected as they fall throughout the growing season. Unless the fallen leaves are collected, up to 40% of the above-ground biological yield is lost (7).

Collecting the fallen leaves is impractical from a breeding stand-point. A simpler approach would be to evaluate genotypes for harvest index at maturity and use these estimates of apparent harvest index, if possible, to screen strains for efficiency. The apparent harvest index would permit screening a larger number of genotypes, but it may or may not adequately reflect the genotype's actual harvest index.

The objectives of this study were to: i) characterize the relationship between the actual and apparent harvest indices, ii) examine their association with other plant characteristics, and iii) evaluate the relative stability of harvest index in soybeans.

MATERIALS AND METHODS

Strains used in this study were selected on the basis of maturity, growth habit, plant height, and yielding ability. Genotypes relatively well adapted to the central Indiana environment were chosen to represent a diverse range in the above characteristics. A listing of the strains used in the experiment along with their parentage, maturity group and growth habit is shown in Table 1.

The experiment was conducted in 1975 and 1976 on the Purdue Agronomy Farm at Lafayette, Indiana. The strains were planted in a Chalmers silty, clay loam soil possessing good fertility on 8 May 1975 and 11 May 1976. Individual plots consisted of three rows, 2.4 m long and spaced 75 cm apart. Plots were overplanted and then thinned to 20 plants/m of row at the first trifoliate stage of development.

Plots were arranged in a randomized, complete block design with three replications. The only restriction on the randomization of entries was that cultivars in adjacent plots did not differ by more than 4 days in maturity.

The 1975 growing season began with topsoil and subsoil moisture in adequate to surplus supply. Rainfall was above normal in June and August, but 5 cm below normal throughout July. In general, air and soil temperatures for the season were slightly above normal.

Soil moisture was in short to adequate supply in the spring of 1976. This situation was relieved in June when rainfall was 8 cm above normal. From midsummer through the remainder of the growing season rainfall and air temperatures were below normal.

A 1.2-m section of the middle row of each plot was enclosed with a 1.9-cm open-mesh, polypropylene netting approximately 1½ months after emergence to retain the fallen leaves. Leaves

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Table 1. Parentage, maturity group, and growth habit of soybean strains used in the experiment.

Strain	Parentage	Maturity group	Growth habit
Hark	Hawkeye × Harosoy	I	I†
Hodgson	Corsoy × [(Lincoln × Richland) × PI 180.501]	I	I
Steele	Blackhawk × Harosoy	I	I
Amsoy 71	Amsoy ^a × (Blackhawk × Harosoy)	II	I
Corsoy	Harosoy × Capital	II	I
Wells	(Harosoy × Kent) × (Blackhawk × Harosoy)	II	I
CX509-19-1-1	Beeson × (Harosoy × T207‡)	II	SD
CX511-117-3-2	Calland × (Harosoy × T207‡)	II	SD
CX520-63	Amsoy 71 × (Harosoy × T207‡)	II	SD
Calland	(Blackhawk × Harosoy) × Kent	III	I
Williams	Wayne × [(Lincoln × Richland) × Adams]	III	I
Woodworth	Wayne × [(Lincoln × Richland) × Adams]	III	I
CX510-369	(Adelphia × Mukden) × (Harosoy × T207‡)	III	SD
CX511-303-2-2	Calland × (Harosoy × T207‡)	III	SD
CX511-171	Calland × (Harosoy × T207‡)	III	SD
Bonus	(Harosoy × Kent) × (Blackhawk × Harosoy)	IV	I
Cutler 71	Cutler × Kent-Rps rxp (SL5)	IV	I
Kent	Lincoln × Ogden	IV	I
S7	(Clark × T117) × [Wayne × (Clark × Kanrich)]	IV	D
CX512-182-3-4	Bonus × (Harosoy × T207‡)	IV	SD
CX512-241-3-1	Bonus × (Harosoy × T207‡)	IV	SD
CX441-181-1	[(Wabash × Hawkeye) × (Hawkeye × Lee)] × Beeson	IV	D
CX454-414-4-2-1	Amsoy × PI 80837	IV	D
CX464-288	Cutler × PI 80837	IV	D

† I, SD and D represent indeterminate, semideterminate and determinate, respectively. ‡ Source of genetic type collection strains in Bernard and Weiss (2).

were collected from each plot at least twice weekly during intense leaf fall. If the leaves were damp when collected, they were placed in a forced-air dryer overnight. All collected leaves were stored in paper bags until weighed.

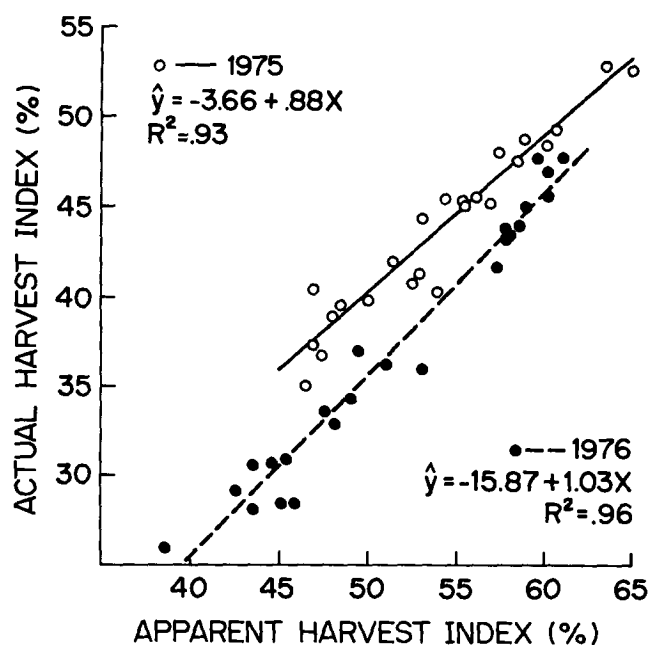
The following five characteristics were measured on each 1.2-m plot: i) maturity date, when 95% of the pods were brown; ii) height (cm) of the plants at maturity; iii) air-dried above-ground mature plant weight (g); iv) air-dried leaf weight (g); and v) air-dried seed weight (g) at harvest.

Six additional characteristics were calculated for each plot from the measured variables listed above: i) biological yield, the total weight of the above-ground dry matter produced by the plants; ii) vegetative yield, the weight of the above-ground, non-grain dry matter; iii) straw weight, the weight of the above-ground stem and pods, not including the seeds; iv) actual harvest index, the ratio of seed yield to biological yield; v) apparent harvest index, the ratio of seed yield to mature plant weight; and vi) leaf index, the ratio of leaf weight to biological yield.

Each plot was harvested after the stems and seeds had thoroughly air-dried. Normally, this was about 1 week after maturity. At harvest, plants within the netting were clipped off at ground level, weighed, and threshed on a portable thresher. Seed weights were taken immediately after threshing.

Moisture percentages were determined for the various plant parts to prevent differences in moisture content from biasing the harvest index calculations. Leaf samples were taken each day leaves were collected, and stem and seed samples were taken each day of harvest. Statistical analysis of the moisture samples indicated that in 1975, the different plant parts contained equal amounts of moisture. In 1976, the stems contained approximately 50% more moisture than did the leaf blades, petioles, and seeds. Consequently, stem weights for 1976 were adjusted so that all plant parts contained equal amounts of moisture.

The data were analyzed as a randomized, complete block design. Least significant differences (L.S.D.'s) were used to com-

**Fig. 1. Relationship between the actual and apparent harvest index means of 24 soybean cultivars in 1975 and 1976.**

pare the individual means of each genotype and maturity group averages at the 5% level of probability. Product-moment correlations between all the characteristics of interest were calculated from entry means. Spearman's coefficient of rank correlation (11) was used to estimate the correlations between the rankings. Simple linear regression was used to examine the relationship between the apparent and actual harvest indices.

RESULTS AND DISCUSSION

The relationship between the actual and apparent harvest indices is shown in Figure 1. The linear regression of the actual harvest index on the apparent harvest index for both the 1975 and 1976 data resulted in highly significant coefficients of determination. The correlations between the rankings of the actual and apparent harvest index means of 0.96 and 0.97 for 1975 and 1976, respectively, were also significant at the 1% level of probability.

Although the slopes of the regression lines were not significantly different, the association between these two characteristics did fluctuate from 1975 to 1976. In 1975, the apparent harvest index means tended to be 23% larger than the actual harvest index values. In 1976, this difference increased to 39%. The change in the leaves' contribution to the above-ground biological yield (leaf index) accounted for the discrepancy. Leaf material comprised a larger portion of the biological yield in 1976 than in 1975 (Tables 2 and 3). Consequently, the difference between the two efficiency indices was greater the second growing season.

The magnitude of the harvest index values also fluctuated from one year to the next. The actual harvest indices tended to be lower in 1976 than in 1975 for two reasons: i) reduced seed yields of strains maturing after 9/13, and ii) increased vegetative dry-matter production, primarily in the form of increased leaf weights (Tables 2 and 3). Only cultivars in maturity groups III and IV experienced a general decline in the ap-

Table 2. Means of actual and apparent harvest indices along with several other soybean characteristics in 1975.

Strain	Growth habit†	Actual harvest index	Rank	Apparent harvest index	Rank	Seed yield	Rank	Straw yield	Leaf wt	Leaf index	Maturity	Plant ht
		%		%		g/plot		g/plot		%	date	cm
Hark	I	48.1	6	57.5	7	294	22	217	100	16	9-5	64
Hodgson	I	53.1	1	63.6	2	257	24	146	80	16	8-27	56
Steele	I	52.8	2	65.1	1	302	20	162	109	19	8-31	58
	$\bar{X} =$	51.3		62.1		284		175	96	17	8-31	
Amsoy 71	I	48.9	4	58.8	5	426	3	300	146	17	9-10	79
Corsoy	I	49.6	3	60.7	3	343	17	222	126	18	9-5	69
Wells	I	47.7	8	56.1	9	370	11	291	148	19	9-11	71
CX509-19-1-1	SD	45.5	9	55.4	12	344	15	276	134	18	9-6	61
CX511-117-3-2	SD	44.5	13	53.2	14	300	21	269	111	16	9-8	61
CX520-63	SD	47.7	7	58.5	6	273	23	198	102	18	9-5	58
	$\bar{X} =$	47.0		57.2		342		259	128	18	9-7	
Calland	I	45.5	9	54.6	11	446	1	385	162	17	9-23	94
Williams	I	41.4	15	52.7	15	435	2	387	226	22	9-26	102
Woodworth	I	45.3	11	56.9	8	401	6	310	182	20	9-21	96
CX510-369	SD	48.6	5	60.2	4	379	9	250	149	19	9-23	79
CX511-303-2-2	SD	39.7	20	48.5	19	411	4	443	182	18	9-16	79
CX511-171	SD	40.6	17	46.9	22	397	7	451	135	14	9-18	74
	$\bar{X} =$	43.5		53.3		411		371	173	18	9-22	
Bonus	I	39.8	19	50.0	18	349	13	348	179	20	9-28	114
Cutler 71	I	37.4	22	46.8	23	374	10	439	195	20	10-1	109
Kent	I	39.2	21	48.0	20	348	14	383	165	18	10-3	99
S7	D	42.2	14	51.5	17	384	8	360	160	18	9-28	81
CX512-182-3-4	SD	45.3	11	55.5	10	406	5	328	165	18	9-25	81
CX512-241-3-1	SD	40.9	16	52.5	16	367	12	334	199	22	9-30	91
CX441-181-1	D	35.1	24	46.5	24	307	18	354	214	24	9-28	58
CX454-414-4-2-1	D	40.4	18	53.9	13	304	19	260	188	25	10-3	53
CX464-288	D	37.0	23	47.4	21	344	15	396	202	22	10-3	69
	$\bar{X} =$	39.7		50.2		354		356	185	21	9-30	
L.S.D. (0.05)		4.1		5.1		81		110	46	4	4	4
L.S.D. (0.05)	$\bar{X} =$	8.6		11.1		110		163	66	7		

† I, SD and D represent indeterminate, semideterminate and determinate, respectively.

Table 3. Means of actual and apparent harvest indices along with several other soybean characteristics in 1976.

Strain	Growth habit†	Actual harvest index	Rank	Apparent harvest index	Rank	Seed yield	Rank	Straw yield	Leaf wt	Leaf index	Maturity	Plant ht
		%		%		g/plot		g/plot		%	date	cm
Hark	I†	43.6	8	58.1	7	353	7	254	203	25	9-5	84
Hodgson	I	47.9	1	61.1	1	372	3	236	167	22	9-1	81
Steele	I	45.1	5	59.0	5	355	6	246	185	24	9-2	86
	$\bar{X} =$	45.5		59.4		360		245	185	24	9-3	
Amsoy 71	I	43.5	9	58.0	8	374	2	271	215	25	9-9	99
Corsoy	I	47.2	3	60.2	2	368	4	245	170	22	9-7	86
Wells	I	44.2	6	58.6	6	342	10	241	191	25	9-7	89
CX509-19-1-1	SD	47.9	1	59.6	4	346	8	235	145	20	9-8	79
CX511-117-3-2	SD	43.9	7	57.7	9	344	9	252	187	24	9-9	84
CS520-63	SD	45.8	4	60.2	2	382	1	253	200	24	9-6	89
	$\bar{X} =$	45.4		59.0		359		250	185	23	9-8	
Calland	I	37.1	11	49.5	13	309	15	315	207	25	9-16	104
Williams	I	34.4	14	49.1	14	336	11	348	291	30	9-20	109
Woodworth	I	36.3	12	51.0	12	325	13	312	259	29	9-15	104
CS510-369	SD	33.0	16	48.1	15	335	12	363	300	30	9-20	109
CX511-303-2-2	SD	41.8	10	57.3	10	358	5	271	231	27	9-15	94
CX511-171	SD	36.1	13	53.1	11	309	15	273	274	32	9-13	91
	$\bar{X} =$	36.4		51.4		329		314	260	29	9-17	
Bonus	I	30.5	19	43.5	21	258	18	333	250	30	9-21	122
Cutler 71	I	30.7	18	44.6	20	279	17	369	299	31	9-21	122
Kent	I	28.1	23	43.5	21	243	23	316	312	36	9-24	109
S7	D	30.8	17	45.4	18	256	19	308	266	32	9-21	96
CX512-182-3-4	SD	33.7	15	47.6	16	323	14	356	282	29	9-20	99
CX512-241-3-1	SD	29.2	20	42.6	23	247	21	333	266	31	9-25	114
CX441-181-1	D	25.9	24	38.6	24	230	24	365	291	33	9-28	96
CX454-414-4-2-1	D	28.4	21	45.9	17	255	20	300	341	38	10-1	76
CS464-288	D	28.4	21	45.1	19	244	22	297	319	37	9-23	91
	$\bar{X} =$	29.5		44.1		262		331	292	33	9-24	
L.S.D. (0.05)		3.5		3.3		54		56	56	5	2	3
L.S.D. (0.05)	$\bar{X} =$	7.0		7.2		67		72	85	8		

† I, SD and D represent indeterminate, semi-determinate and determinate, respectively.

Table 4. Correlations between 10 soybean characteristics for 1975 (right diagonal) and 1976 (left diagonal), (n = 24).

	Apparent harvest index	Actual harvest index	Seed yield	Biological yield	Vegetative yield	Mature plant wt	Straw wt	Leaf wt	Maturity	Plant ht
Apparent harvest index		0.96**	-0.28	-0.74**	-0.86**	-0.71**	-0.88**	-0.67**	-0.73**	-0.40*
Actual harvest index	0.98**		-0.23	-0.72**	-0.86**	-0.64**	-0.82**	-0.80**	-0.82**	-0.38*
Seed yield	0.91**	0.91**		0.83**	0.67**	0.87**	0.67**	0.52**	0.42*	0.65**
Biological yield	-0.60**	-0.64**	-0.27		0.97**	0.98**	0.95**	0.82**	0.74**	0.69**
Vegetative yield	-0.90**	-0.92**	-0.72**	0.83**		0.93**	0.97**	0.86**	0.80**	0.64**
Mature plant wt	0.12	0.15	0.51**	0.60**	0.16		0.95*	0.70**	0.66**	0.68**
Straw wt	-0.88**	-0.84**	-0.60**	0.85**	0.92**	0.38*		0.72**	0.71**	0.62**
Leaf wt	-0.84**	-0.92**	-0.72**	0.80**	0.93**	0.00	0.78**		0.86**	0.54**
Maturity	-0.94**	-0.95**	-0.86**	0.63**	0.92**	-0.12	0.81**	0.89**		0.57**
Plant ht	-0.63**	-0.58**	-0.39*	0.61**	0.62**	0.39*	0.78**	0.47*	0.50**	

*, ** Indicate significance of the 5 and 1% levels of probability, respectively.

parent harvest index values in 1976. The decrease was attributed to the reduced seed yields of the later-maturing entries.

Varying environmental conditions affected the phenotypic expression of the actual and apparent harvest indices as well as altering the relationship between these two ratios. Nevertheless, the correlation between the actual and apparent harvest indices remained highly significant under two environmental conditions and over a range of genotypes that possessed significantly different leaf indices. Therefore, measurement of the apparent harvest index should permit a valid comparison of the relative efficiency of a group of genotypes.

Evaluating genotypes on the basis of apparent harvest index may alter the breeding strategy of attempting to combine the superior efficiency of one genotype with the high biological yield of another. Since biological yield is eliminated as a criterion for evaluation when the apparent harvest index is used, emphasis must be placed on evaluating strains for seed yield coupled with relative seed-yield efficiency.

The correlations derived from the 1975 and 1976 data are listed in Table 4. In general, the actual and apparent harvest indices were negatively correlated with biological and vegetative yields, straw and leaf weights, maturity, and plant height. The latter six characteristics were positively correlated with each other. In contrast to these relatively uniform relationships, only mature plant weight was consistently correlated with seed yield for the 2 years. This result was expected considering the components of mature plant weight. Previous reports have found that, generally, maturity and plant height were positively correlated with seed yield (1, 4, 5, 8, 9). Buzzell and Buttery (3) reported that harvest index was negatively correlated with plant height, maturity, straw weight, and seed yield.

The consistent as well as inconsistent associations can be analyzed by examining each characteristic's relationship with date of maturity. In 1975, environmental conditions slightly favored the later-maturing entries. Consequently, seed yield and maturity were positively correlated. Biological and vegetative yields along with the components of vegetative yield, straw weight, and leaf weight, were also positively correlated with maturity. The increase in biological yield in the late-maturing entries was partially due to higher seed yields, but the majority of the increase came from the

production of additional vegetative dry matter (Tables 2 and 3). Because the increase in vegetative yield was larger than the corresponding increase in seed yield for the late-maturing entries, the harvest index ratios declined progressively as maturity lengthened. Harvest index and seed yield were not correlated in 1975.

In 1976, environmental conditions favored the early-maturing entries, resulting in an inverse relationship between seed yield and maturity. Although seed yields of the late entries were reduced, biological yield and vegetative yield remained positively correlated with maturity. Under these conditions the inverse relationship between maturity and the harvest indices became stronger and seed yield was positively correlated with the harvest indices.

It appears that even under environmental conditions favorable to the late-maturing entries as in 1975, an inverse relationship will exist between harvest index and maturity. In general, the photoperiod response of the late-maturing strains lengthened the vegetative and reproduction dry matter accumulation periods in comparison with earlier entries. However, the latter portion of the grain-filling period for the late-maturing genotypes generally occurs when air temperatures and soil moisture begin to decline. These environmental conditions may restrict late-maturing genotypes from utilizing the vegetative material as effectively as earlier-maturing genotypes. Considering the magnitude and consistency of the association between harvest index and maturity, comparisons of efficiency should be made among genotypes with similar maturities.

The association between the harvest indices and growth habit can be characterized by examining Tables 2 and 3. Comparisons of the actual and apparent harvest index means among the entries with similar maturities showed that no consistent relationship existed between a genotype's growth habit and its harvest index.

If selection among strains is to be effective, the relative performance of a group of genotypes must remain somewhat stable, even when the genotypes are subjected to a variety of environmental conditions. To examine the relative performance of the strains used in this study, correlations were calculated between the 1975 and 1976 means for each characteristic. These correlations, recorded in Table 5, showed that no relationship existed between the 1975 and 1976 seed yield means. Except for mature plant weight, correlations for the remaining characteristics were all

Table 5. Correlations between 1975 and 1976 means for 10 soybean characteristics, (n = 24).

Characteristic	Correlation coefficient
Apparent harvest index	0.66**
Actual harvest index	0.78**
Seed yield	-0.05
Biological yield	0.54**
Vegetative yield	0.69**
Mature plant wt	0.13
Straw wt	0.53**
Leaf wt	0.74**
Maturity	0.96**
Plant ht	0.92**

** Indicates significance at the 1% level of probability.

highly significant. These correlations indicate that the relative performance of the entries for seed yield was drastically changed from one season to the next. In contrast, maturity and plant height were two traits whose relative performance was more consistent in the different environments. The stability of the harvest index characteristics fell within these two extremes. The relative performance of the entries based on their harvest index means did shift to some extent from 1975 to 1976, but the magnitude of the changes in relative performance was much smaller than the observed changes in seed yield.

Considering the strong relationship between harvest index and maturity, sufficient genetic variability must be present among genotypes possessing similar maturities for selection and evaluation based on harvest index estimates to be meaningful. Although harvest index and seed yield were not consistently correlated, the limited genotypic sample within each maturity group prevented an extensive evaluation of their relationship in a narrow maturity range. Future experiments designed to evaluate harvest index may be somewhat less restricted, because determining a genotype's apparent harvest index is easier than measuring the actual harvest index. This should allow for a more

complete and accurate characterization of the variability of harvest index in soybeans and its usefulness as a plant breeding tool.

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